



When will the immune-stimulating antibody conjugates (ISACs) be transferred from bench to bedside?

Chen Fu^{a,b,1}, Weiwei Tong^{c,1}, Lifeng Yu^{a,b}, Yuxi Miao^{a,b}, Qian Wei^{a,b}, Zhaojin Yu^{a,b,*},
Bo Chen^{d,**}, Minjie Wei^{a,b,***}

^a Department of Pharmacology, School of Pharmacy, China Medical University, Shenyang 110122, PR China

^b Liaoning Key Laboratory of Molecular Targeted Anti-tumor Drug Development and Evaluation, China Medical University, Shenyang 110122, PR China

^c Department of Laboratory Medicine, Shengjing Hospital of China Medical University, Shenyang 110122, PR China

^d Department of Breast Surgery, The First Hospital of China Medical University, No. 155, Nanjing North Street, Shenyang 110122, PR China

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ABSTRACT

Immunostimulatory antibody conjugates (ISACs) as a promising new generation of targeted therapeutic antibody-drug conjugates (ADCs), that not only activate innate immunity but also stimulate adaptive immunity, providing a dual therapeutic effect to eliminate tumor cells. However, several ISACs are still in the early stages of clinical development or have already failed. Therefore, it is crucial to design ISACs more effectively to overcome their limitations, including high toxicity, strong immunogenicity, long development time, and poor pharmacokinetics. This review aims to summarize the composition and function of ISACs, incorporating current design considerations and ongoing clinical trials. Additionally, the review delves into the current issues with ISACs and potential solutions, such as adjusting the drug-antibody ratio (DAR) to improve the bioavailability of ISACs. By leveraging the affinity and bioavailability-enhancing properties of bispecific antibodies, the utility between antibodies and immunostimulatory agents can be balanced. Commonly used immunostimulatory agents may induce systemic immune reactions, and BTK (Bruton's tyrosine kinase) inhibitors can regulate immunogenicity. Finally, the concept of grafting ADC's therapeutic principles is simple, but the combination of payload, linker, and targeted functional molecules is not a simple permutation and combination problem. The development of conjugate drugs faces more complex pharmacological and toxicological issues. Standing on the shoulders of ADC, the development and application scenarios of ISAC are endowed with broader space.

Abbreviations: ACC, Adrenocortical carcinoma; ADC, antibody-drug conjugate; ADCP, Antibody-Dependent Cellular Phagocytosis; AE, Adverse Event; ALL, Acute Lymphocytic Leukemia; AML, Acute Myelocytic Leukemia; APC, antigen-presenting cell; BCMA, B cell maturation antigen; BCR, B-cell receptor; BTK, Bruton's tyrosine kinase; CD8+, Cluster of differentiation 8 positive; CD74, Cluster of differentiation 74; CDC, conventional dendritic cell; CEA, carcinoembryonic antigen; CLL, Chronic Lymphocytic Leukemia; COVID-19, Corona Virus Disease 2019; CR, chemokine receptor; CRS, cytokine release syndrome; CTCL, cutaneous T cell lymphoma; DAR, Drug-to-Antibody Ratio; DC, dendritic cell; DCR, Disease Control Rate; DLT, dose-limiting toxicity; EC50, Half maximal effective concentration; EGFR, epidermal growth factor receptor; EMSO, European Society for Medical Oncology; ES-SCLC, extensive-stage small cell lung cancer; HCC, hepatocellular carcinoma; HER2, human epidermal growth factor receptor-2; HNSCC, Head and Neck Squamous Cell Carcinoma; IFNs, interferons; IgG, Immunoglobulin G; IO, Immuno-oncology; IrAE, immune-related adverse event; ISAC, Immune-stimulating antibody conjugate; LPS, Lipopolysaccharide; MAbs, monoclonal antibodies; MCL, Mast cell leukemia; MDC, monocyte-derived dendritic cell; MDR, multidrug-resistant; MIBC, Muscle-invasive bladder cancer; MOA, Mechanism of action; Mpk, milligrams per kilograms; MTD, Maximum Tolerated Dose; NBCC, Nodular Basal Cell Carcinoma; NSCLC, non-small cell lung cancer; ORR, Objective Response Rate; PBMC, Peripheral blood mononuclear cell; PD, progressive disease; PD-1, Programmed cell Death 1; PD-L1, Programmed cell Death-Ligand 1; PDGFR- α , platelet-derived growth factor receptor alpha; PK, Pharmacokinetics; PR, partial response; PRRs, pattern recognition receptors; QW, quaque week; Q2W, quaque 2 week; Q3W, quaque 3 week; QSP, Quantitative Systems Pharmacology; R&D, research and development; RO, receptor occupancy; RP2D, Recommended Phase II Dose; SD, stable disease; SITC, The Society for Immunotherapy of Cancer; SLAMF7, Signaling Lymphocytic Activation Molecule Family, Member 7; STING, Stimulator of Interferon Genes; TAA, tumor-associated antigens; TCR, T-cell receptor; TLR, Toll-Like Receptor; TNF, tumor necrosis factor; VEGF, vascular endothelial growth factor.

* Corresponding author at: Department of Pharmacology, School of Pharmacy, China Medical University, Shenyang 110122, PR China.

** Corresponding author.

*** Correspondence to: Liaoning Medical Diagnosis and Treatment Center, Shenyang 110000, PR China.

E-mail addresses: yuzhaojin19830813@163.com (Z. Yu), bochen@cmu.edu.cn (B. Chen), mjwei@cmu.edu.cn (M. Wei).

¹ These authors have contributed equally to this work and share the first authorship

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1. Introduction

Antibody-drug conjugates (ADCs) for precise tumor targeting are no longer a challenge[1,2]. Researchers have now developed immune-stimulating antibody conjugates (ISACs) based on the ADC design concept. ISAC is specifically engineered for the covalent attachment of an immune agonist to a tumor-targeting antibody. This innovative methodology leverages the antibody as a vehicle for precise and localized delivery of the immune agonist to the tumor microenvironment. Such a strategy addresses the inherent challenges associated with the systemic administration of immune stimulants, ultimately enhancing the efficacy of anti-tumor therapy, all while mitigating the potential for systemic side effects. The ISAC consists of three main components: tumor-targeting antibodies, non-degradable linkers, and a specific immunostimulant that activates the innate immune system[3] (Fig. 1A). ISACs are a more successful form of ADCs and act as a "spy" that infiltrates inside the tumor and destroys it.

The ISAC functions through three main steps: antigen recognition, Fc-mediated endocytosis in antigen-presenting cells (APCs), and activation of toll-like receptors (TLRs) in APC cells[4]. Structurally, ISACs are similar to ADCs, as they both consist of three components. However, ISACs contain innate immune agonists or modulators instead of toxins. Although structurally similar, their mechanisms of action are completely different[5] (Fig. 1B). Conventional ADC designs use antibodies to deliver toxins into tumor cells to kill them, which can be understood as a mechanism of antibody-conjugate drugs[6]. ISACs are a novel type of immunotherapy that involves conjugating innate immune agonists or modulators with macromolecular antibodies to deliver them to the tumor microenvironment. This activation of T cells enhances the killing effect of tumor immunity[7]. ISACs can generate new immune responses by activating and recruiting bone marrow cells. Then, the activated bone marrow cells start a feed-forward cycle through chemical signals by releasing cytokines and chemokines, attracting other immune cells, and lowering the activation threshold of the immune response. This reprograms the tumor microenvironment and triggers the adaptive immune response against the tumor, which provides a durable response to cancer patients[8]. ISAC combines the precision of antibody targeting with the strength of the innate and adaptive immune systems to target a single marker on the surface of tumor cells[9]. Antibodies are responsible for precise tumor targeting, while agonists activate innate and adaptive immunity, enabling the transformation of a "cold tumor" into a "hot tumor." [10]. Theoretically, ISACs have the potential to be used in various ways and could have a potent tumor-killing effect. However, the risk is also relatively high, as there is a possibility of activating the systemic non-specific immune response, and in severe cases, cytokine storms can be life-threatening.

2. The ISAC components

2.1. Antibodies in the ISAC

Antibodies serve as targeted agents in ISAC. The majority of approved ADC products currently available consist of monoclonal antibodies that are either humanized or fully human. ISAC are engineered monoclonal antibodies conjugated with ligands of synthetic pattern recognition receptors (PRRs). The successful development of antibodies for ISAC requires the TSA(tumor specific antigen) or TAA(tumor associated antigen)[11]. Different types of antibodies exist, including IgG1, IgG2, IgG3, and IgG4, but IgG1 is still the dominant antibody used in ISACs. The choice of IgG1 is based on two factors. Firstly, the affinity of the antibody for the FcγR is important because FcγR is expressed as a receptor in immune cells[12,13]. The greater affinity makes FcγR easier to recruit immune cells to kill the tumor. Secondly, if the antibody can be more stable and have a longer half-life, it has more chance to kill the tumor[14].

2.1.1. Commonly used antibodies and targets for ISAC

Bolt Therapeutics has developed a technology platform called Bolt-body™, which has been patented in many countries around the world. Several monoclonal antibodies have been tested by this platform, and all of them have demonstrated excellent tumor inhibition effects[15]. The antibody component of ISAC remains the same as for conventional ADCs. Antibodies (corresponding target) usually used in ISACs are pembrolizumab (PD-1), nivolumab (PD-1), ipilimumab (CTLA-4), atezolizumab (PD-1), trastuzumab (HER2), cetuximab (EGFR), rituximab (CD20), pertuzumab (HER2), bevacizumab (VEGF), etanercept (TNF), olaratumab (PDGFR-α), elotuzumab (SLAMF7) and so on[16].

2.1.2. Considerations for antibody selection

The selection of antigens is the most primary and crucial step in the development of an ISAC. The appropriate selection of ISAC antigens must meet the following criteria: high expression in tumors, low or no expression in healthy tissues, overexpression of the antigen on the surface of tumor cells, promotion of tumor growth and survival, internalization characteristics to facilitate the transport of the ISAC into the cells for toxic effects, and specific binding ability of the antibody (the "A" in ISAC) to accurately target the cells and deliver the toxin[17].

Additionally, when designing antibodies, it is important to consider the affinity of the antibody for the target antigen. Antibodies with high affinity can ensure better accumulation of toxins within tumor cells, but they also have faster rates of internalization and degradation, which limits their ability to penetrate tumors and reduces their cytotoxicity towards tumor cells located at the distal end of blood vessels[18]. On the other hand, antibodies with low affinity have slower rates of internalization and degradation, allowing them to remain on the surface of tumor cells for a longer time. This enables them to penetrate tumor cells, resulting in stronger anti-tumor activity[19]. However, it is important to note that high-affinity antibodies do not necessarily equate to high efficacy[20]. Antibodies with high affinity for their target may be rapidly eliminated from circulation due to rapid internalization upon binding. High affinity may also reduce the ability of antibodies to penetrate deep into tumors, as a result of tight binding or rapid internalization upon encountering the first cell, a phenomenon known as the binding site barrier (BSB)[21].

Based on simulation and experimental analysis, there is a delicate balance between the molecular size and affinity of therapeutic antibodies and the extent of tumor retention and penetration. Treatment regimens of moderate size (~25 kDa) exhibit the lowest tumor uptake, while both extremely small (<20 kDa) and large (>100 kDa) treatment regimens demonstrate higher tumor uptake and retention rates[22,23].

Species cross-reactivity is an important consideration in many antibody development techniques, particularly in complex toxicology studies where it is crucial for ISAC development[24]. Aligning the target amino acid sequence with its homologous sequences in other species helps determine the homology or consistency between humans and other species (such as non-human primates or rodents) as targets[25]. Sequence identity can roughly predict the likelihood of cross-reactivity of antibodies (higher sequence identity between the antigen and the antibody increases the probability of cross-reactivity). Conversely, generating cross-reactive antibodies is challenging. Therefore, when faced with multiple potential targets from omics studies, it is advantageous to prioritize targets with high sequence homology[26]. However, linear sequence homology does not always translate into the three-dimensional structure of the protein. For instance, despite low overall homology in other regions of the protein, ligand-binding sites may still be well conserved across species[27]. Therefore, antibodies that bind to homologs of human targets in non-human primates or rodents are typically screened. Lastly, since the ability of cells to internalize antibodies is a critical factor in their effectiveness, evaluating the internalization capability of the target and subsequently delivering the antibody and payload to the appropriate intracellular compartment (usually lysosomes) is crucial[28]. Therefore, the selection of

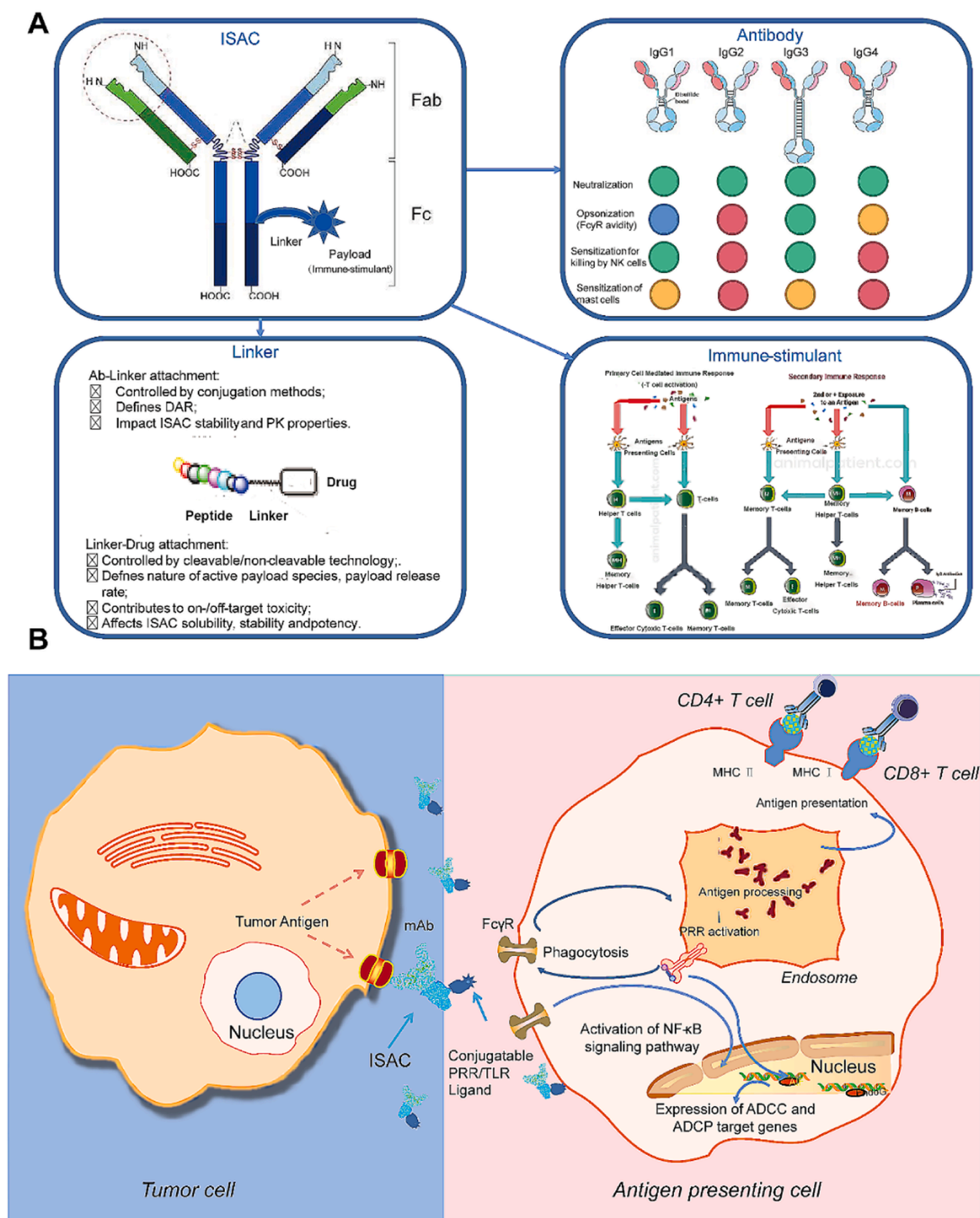


Fig. 1. ISAC structure and mechanisms. (A) Diagram of the ISAC structure, (B) The mechanism of action of ISAC.

appropriate antibodies for ISAC is a challenging process that requires careful consideration of numerous factors[29].

2.2. Linkers in the ISAC

The linker is an essential component in ISAC, as it covalently links antibodies to immune stimulants. The key requirement for the linker is to ensure the chemical stability of ISAC in the bloodstream and allow for rapid release of the payload at the target site after internalization[30]. Additionally, the linker's appropriate hydrophilicity/lipophilicity can enhance payload conjugating and reduce immunogenicity, which is also a crucial aspect of the linker[31].

Linkers can be classified into two categories: cleavable linkers and non-cleavable linkers. Cleavable linkers release payloads through physiological conditions in the cell and can be further subdivided into acid-sensitive, protease-sensitive, or glutathione-sensitive cleavable linkers depending on the intracellular environment[32,33]. Non-cleavable linkers, on the other hand, can only be degraded by the lysosomes of the cells. As a result, non-cleavable linkers used in ISACs have a longer half-life and lower off-target toxicity in the blood than cleavable linkers. Therefore, most ISAC drugs currently use non-cleavable linkers.

Recent advancements in linker chemistry have expanded the functionality of linkers beyond just serving as a bridge between antibody and payload. Modified linkers can now promote ISAC activation and alter the polarity or charge of the final metabolite, leading to better activity against multidrug-resistant (MDR) cells due to better payload retention within the cell[34]. Some linker designs can also induce bystander effects, resulting in greater anti-tumor activity in vivo. Linkers are chemical structures that link payload loads to monoclonal antibodies (mAbs)[35]. The covalent linker needs to be stable in the circulatory system and readily hydrolyzable after endocytosis by the target cells to release the active component. The linker can affect the pharmacodynamic and pharmacokinetic properties of the ISAC to some extent[36].

2.3. Drugs in the ISAC

ISAC is an emerging field and the drugs used are mostly TLR7, TLR8, TLR9 agonists, or STING agonists. Some testing has also been done for other members of the TLR family or any combination (e.g. TLR7/8 dual activation)[37]. Currently, the main ISAC drugs involved include the TLR agonist SBT6050, SBT6290, and BDC-1001; the STING agonist XMT-2056, and the Treg cell regulatory modulators ADCT-301. However, many of these drugs are also defined as ADC by the companies themselves, perhaps it is also because there is not much difference between the appearance of the drug and the technique[38]. The representative ISAC, BDC-1001, is currently in Phase II clinical trials in patients with HER2-expressing tumors. In addition to the fastest progressing BDC-1001 targeting HER2, ISAC targeting PD-L1 and CEA is also under development. The ISAC targeting PD-L1 achieved essentially complete inhibition when used in C57BL/6 mice at 5mpk (milligrams per kilogram). The antibody to the ISAC partially targets the tumor cells and the TLR agonist carried by the antibody will activate the tumor surrounding DC cells, using the body's innate immune system to kill tumor cells. The Boltbody™ platform technology reinvigorates the immune system of both dendritic cells and other myeloid cells generate effective antitumor immunity[39]. Boltbody™ ISAC is a unique class of immuno-oncology (IO) therapies, its function is expected to generate systemic immune memory with lasting clinical benefit[15].

Currently, TLR agonists have potential therapeutic promise as one of the research frontier areas in immunotherapy. TLRs are evolutionarily conserved innate immune molecules and also serve as a bridge between innate and adaptive immunity[40]. IRFs (interferon-regulatory factors), whose signaling results in the production of mainly pro-inflammatory cytokines, and type I interferons (IFNs)[41]. Several drugs targeting TLRs have been developed and several have been marketed, with TLR7,

TLR8, and TLR9 in the clinical phase progressing relatively quickly [42–44](Table 1). Innate immune agonists may be able to promote the initiation and recruitment of tumor-specific cells and are therefore highly promising cancer immunotherapy (Fig. 2A).

Based on the important role of STING in promoting anti-tumor immunity, several teams have initiated the development of STING agonists (Table 2), which may have the potential to trigger innate immune activation and lead to an adaptive immune response that controls tumor growth when used as monotherapy or in combination with checkpoint blockade therapies (Fig. 2B)[89–91]. The TLRs and STING drugs have the potential to become immune agonists as part of the ISAC. Combination therapies targeting TLRs and STING drugs with tumor vaccines, immune checkpoint inhibitors, etc. will also be an important area of research.

In summary, TLRs are a type of innate immune receptor responsible for directly or indirectly detecting pathogen-associated molecular patterns (PAMPs) and initiating responses through activation of innate and adaptive immune pathways. STING participates in cytosolic DNA virus recognition and innate immune function through the cGAS-STING pathway, using cGAMP as a second messenger. Both TLR agonists and STING agonists have the potential to reshape the tumor immune microenvironment, enhance tumor immunotherapy responsiveness and efficacy, addressing therapeutic challenges that traditional ADC drugs may not resolve.

3. ISAC development and clinical applications

BDC-1001 is the first ISAC based on the Boltbody™ platform, consisting of a trastuzumab biosimilar coupled to Bolt's proprietary TLR7/8 dual agonist by a non-cleavable linker for the treatment of HER2+ positive solid tumors. TLR7/8 are important proteins involved in innate immunity and serve as a bridge between non-specific and specific immunity. BDC-1001 stimulates anti-tumor activity in a three-step response. In the first step, BDC-1001 enters the body using a trastuzumab analog that specifically binds to HER2-expressing tumor cells and kills them. The second step involves activating innate immunity. When binding to tumor cells, the Fc segment of the IgG1 antibody binds to the Fc receptor (FcγR), triggering phagocytosis by APCs, our innate immunity, to kill the tumor cells. At the same time, the TLR7/8 agonist attached to BDC-1001 is phagocytosed into the lysosome, and the tumor antigen and TLR7/8 agonist are delivered to the bone marrow cells (leucocytes, red blood cells, platelets, etc.), activating the myeloid APCs. The third step involves mediating acquired immunity. The TLR7/8 agonist awakens the myeloid APCs to present the TAA to the T cells. Tumor cells containing this antigen are seen as "cohorts" acquired immunity is triggered, and T cells are mobilized to hunt down and kill tumor cells. This also transforms a cold tumor, which is not immune-responsive, into a hot tumor, which is immune-responsive[107]. Only when all three steps are fulfilled can the innate and adaptive immune response be regulated to avoid unexpected systemic immune stimulation (Fig. 3).

On December 6, 2021, Bolt Therapeutics presented Phase I/II clinical data for BDC-1001 at the ESMO meeting. 1 PR (partial remission) and 12 SD (stable disease) out of 40 evaluable patients, with an overall remission rate ORR of only 2.5% and a disease control rate DCR of 32.5%. This is not the first ISAC that failed. 3 months ago, Silverback Therapeutics presented interim Phase I/Ib clinical data results for its new immune agonistic ADC drug SBT6050 at the ESMO meeting, with 1 PR, 3 SD, 10 PD, and an ORR (overall remission rate) of only 7% in 14 evaluable patients[108]. On 31 March 2022, Silverback Therapeutics announced its 2021 financial results and future strategic deployment, notably the axing of two of the company's heavyweight ISAC pipelines, announcing the termination of the SBT6050 program[109], along with the abandonment of SBT6290. These results highlight the challenges of developing ISACs and the need for further research to improve their efficacy. It is important to note that clinical trial results are not always

Table 1
TLR7, TLR8, TLR7/8, TLR9 agonists for cancer patients in clinical trials.

Target (s)	Agonist	Indication(s)	Phase (s)	Status	Route	Notes	Clinical Number	Ref
TLR7	852 A	ALL AML CLL Lymphoma Multiple myeloma	II	Terminated	s.c	As a single agent	NCT00276159	[45]
		Breast cancer Cervical cancer Endometrial cancer Ovarian cancer	II	Completed	s.c.	As a single agent	NCT00319748	[46]
		Melanoma	II	Completed	i.v	As a single agent	NCT00189332	[47]
		Melanoma	I	Completed	Topical	As a single agent	NCT00091689	[48]
		Chemorefractory solid tumors	I	Completed	i.v	As a single agent	NCT00095160	[49]
	BNT411	ES-SCLC Solid tumors	I/II	Not yet recruiting	i.v	As a single agent or combined with atezolizumab and chemotherapy	NCT04101357	[50]
	DSP-0509	Advanced solid tumors	I/II	Recruiting	i.v	As a single agent or combined with pembrolizumab	NCT03416335	[51]
	LHC165	Advanced solid tumors	I	Recruiting	i.t.	As a single agent or combined with PDR001	NCT03301896	[52]
		Advanced HER2 ⁺ solid tumors	I	Recruiting	i.v.	As a single agent	NCT03696771	[53]
		Actinic keratosis	II	Completed	Topical	As a single agent	NCT01583816	[54]
		Brain tumors	II	Active, not recruiting	Topical	Combined with DCs pulsed with autologous tumor cell lysates	NCT01204684	[55]
		CTCL	I/II	Completed	Topical	As a single agent	NCT01676831	[56]
		Melanoma	I	Completed	Topical	Combined with a peptide-based vaccine	NCT01748747	[57]
			I	Completed	Topical	Combined with a peptide-based vaccine	NCT00470379	[58]
			I/II	Active, not recruiting	Topical	Combined with a peptide-based vaccine ± poly-ICLC	NCT02126579	[59]
			II	Active, not recruiting	Topical	Combined with a peptide-based vaccine	NCT00960752	[57]
		MIBC	II	Terminated	Topical	Combined with CDX-1307, chemotherapy, GM-CSF, and poly-ICLC	NCT01094496	[60]
		nBCC	I/II	Terminated	Topical	As a single agent	NCT01808950	[61]
		NY-ESO-1 ⁺ tumors	I/II	Completed	Topical	Combined with CDX-1401 ± poly-ICLC	NCT00948961	[62]
		Advanced tumors	I	Completed	Topical	Combined with a peptide-based vaccine	NCT00821652	[63]
	RO7119929	Biliary tract tumors HCC Hepatic metastases	I	Not yet recruiting	p.o.	As a single agent	NCT04338685	[64]
	TQ-A3334	NSCLC	I/II	Recruiting	p.o.	As a single agent or combined with anlotinib	NCT04273815	[65]
TLR7/ TLR8	BDC-1001	HER2 ⁺ advanced solid tumors	I	Recruiting	n.s.	As a single agent or combined with pembrolizumab	NCT04278144	[66]
	CV8102	ACC HNSCC Melanoma SCC	I/II	Recruiting	n.s.	As a single agent or combined with standard of care PD-1 blockage	NCT03291002	[67]
		HCC	I/II	Completed	i.d.	Combined with cyclophosphamide and a peptide-based vaccine	NCT03203005	[68]
	NKTR-262	Advanced/metastatic solid tumors	I/II	Recruiting	i.t.	Combined with pegylated IL2 ± nivolumab	NCT03435640	[69]
TLR8	Motolimod (VTX-2337)	Advanced solid tumors	I	Terminated	s.c.	Combined with cyclophosphamide and pegylated G-CSF	NCT02650635	
			I	Completed	s.c.	Combined with cetuximab	NCT01334177	[70]
		HNSCC	I	Recruiting	i.t. s.c.	As a single agent or combined with nivolumab	NCT03906526	[71]
			I	Terminated	s.c.	Combined with cetuximab ± nivolumab	NCT02124850	
			I	Not yet recruiting	i.t.	Combined with nivolumab	NCT04272333	[72]
			I	Completed	n.s.	Combined with cetuximab and multimodal chemotherapy	NCT01836029	[73]
		Lymphoma	I	Terminated	i.t.	Combined with radiation therapy	NCT01289210	
		Ovarian cancer	I	Completed	s.c.	Combined with paclitaxel or PLD	NCT01294293	[74]
			I/II	Active, not recruiting	s.c.	Combined with durvalumab and PLD	NCT02431559	[75]
TLR9	DN1508052-01 MGN1703	Advanced Solid Tumors	I	Completed	i.v	As a single agent	NCT03934359	[76]
		Advanced Cancers	I	Active, not recruiting	i.v.	Combined with Ipilimumab	NCT02668770	[77]
		Melanoma						
		Small Cell Lung Cancer	II	Completed	s.c.	As a single agent	NCT02200081	[78]
	SD-101	Advanced Colorectal Carcinoma	II	Completed	i.v. or p. o.	As a single agent	NCT01208194	[79]
		Metastatic Uveal Melanoma in the Liver	I	Recruiting	i.v.	Combined with Ipilimumab	NCT04935229	[80]
		Hepatocellular Carcinoma	I	Recruiting	i.v.	Combined with Ipilimumab/ Pembrolizumab /Nivolumab	NCT05220722	[81]
		Intrahepatic Cholangiocarcinoma						
		B-Cell Non-Hodgkin Lymphoma	I	Active, not recruiting	i.v.	As a single agent	NCT03410901	[82]
		Grade 1 Follicular Lymphoma						

(continued on next page)

Table 1 (continued)

Target (s)	Agonist	Indication(s)	Phase (s)	Status	Route	Notes	Clinical Number	Ref
	IMO-2125	Grade 2 Follicular Lymphoma Grade 3a Follicular Lymphoma Lymphoplasmacytic Lymphoma Mantle Cell Lymphoma Marginal Zone Lymphoma Small Lymphocytic Lymphoma Metastatic Melanoma	III	Terminated (Lack of Efficacy)	i.v.	Combined with Ipilimumab	NCT03445533	[83]
		Refractory Solid Tumors Melanoma	I	Completed	i.t.	As a single agent	NCT03052205	[84]
		Hepatitis C, Treatment Naïve, Genotype 1 Patients	I	Completed	s.c.	As a single agent	NCT00990938	[85]
		Solid Tumor	II	Active, not recruiting	i.v.	Combined with Tilsotolimod/ Nivolumab/Ipilimumab	NCT03865082	[86]
	CMP-001	Squamous Cell Carcinoma of Head and Neck	II	Active, not recruiting	i.t.	Combined with Pembrolizumab	NCT04633278	[87]
		Triple Negative Breast Cancer	II	Active, not recruiting	i.t.	As a single agent	NCT04807192	[88]
		Melanoma Advanced Melanoma Metastatic Melanoma Unresectable Melanoma	II	Active, not recruiting	i.t.	Combined with Nivolumab	NCT04698187	[83]

conclusive, and further studies may reveal different outcomes. However, the lack of success in recent trials underscores the complexity of developing ISAC and the importance of careful selection and design of target antigens, antibodies, and immunostimulants in the drug development process.

Systemic administration of an ISAC targeting HER2 (HER2-ISAC) in preclinical models showed local immune activation leading to robust single-agent activity, anticancer host immune memory, and epitope spreading. Pre-clinical data presented at the SITC(The Society for Immunotherapy of Cancer) 2019 meeting showed the complete regression of trastuzumab-resistant tumors by HER2-ISAC and protection of immune memory against homozygous mice with tumor cells no longer expressing HER2 antigens, which demonstrates a durable and meaningful response to HER2-expressing cancers and the therapeutic potential of HER2-ISAC[110].

Following the publication of the clinical results, Bolt Therapeutics has indicated that it will continue with monotherapy dose escalation and evaluate weekly dosing regimens, with the next step being to initiate a combination therapy strategy with trastuzumab/capecitabine. In addition to these two companies, foreign companies with a presence in this field include Novartis, ADC Therapeutics, Mersana Therapeutics, etc. There are also some Chinese pharmaceutical companies including HENGRUI MEDICINE and Beigene laying out their R&D for ISAC. Innovent Biologics introduced Bolt's platform technology, and Qide Medical Technology chose to collaborate with Brightgene Bio-Medical Technology to develop ISAC. Some ISACs are now entering clinical research (Table 3).

Immunotherapy (IO) has historically focused on T-cell-driven effects, including immuno-stimulatory antibody conjugates that comprise tumor-targeting antibodies conjugated to an immunostimulatory payload[131]. ISACs may require tumor-associated antigens (TAA) and tumor-resident bone marrow cells to be effective. Understanding the situation of T-cells is relevant to cancer biology and drug development. T-cell infiltration of human cancer cells is generally understood as a hallmark of immune recognition, and maximizing the anti-tumor potential of infiltrating T cells is currently the subject of intense research. However, only about 10% of intratumoral CD8(+) T cells recognize autologous tumors, suggesting that therapeutic strategies that increase the frequency and width of tumor-specific TCR (T-cell receptor) may be critical to improving immunotherapy efficacy[132]. Tumor-infiltrating myeloid cells and CD8(+) T cells differ in different indications[133]. For example, intra-tumor CD8(+) T cells < 100/mm² were observed in

CRC, whereas higher CD8(+) T cell numbers were observed in other cancers[134]. In contrast to T cells, substantial myeloid cell infiltration was observed in all tumor types[135]. While MSI (Microsatellite Instability) status was associated with higher CD8(+) T cell infiltration in CRC, myeloid cells were abundant in both MSI and MSS (Microsatellite Instability-Stable) samples[136]. Monocyte-derived DCs (mDC) and conventional DCs (cDC) that responded strongly to TLR activation were present in all indications, with the highest numbers in NSCLC[137,138]. HER2+ myeloids were heavily infiltrated with plasma cell DCs, key TLR7-responsive DCs[139]. PD-L1 expression on immune cells was associated with higher numbers of CD8(+) T cells and myeloid cells [140]. In contrast to HER2 and PD-L1, CEA expression was not associated with immune infiltration[141]. A large number of myeloid infiltrates were observed in solid tumors, independent of T-cell infiltration [142]. The presence of myeloid cells in multiple tumor types provides a wide range of therapeutic targets for ISACs. These therapies can activate the innate immune system and act as a bridge to adaptive immunity. With the development of new IO therapy strategies such as ISACs with TLRs-agonists, targeting myeloid cells may become increasingly relevant to cancer treatment.

Looking at the antibody component of ISAC, the current clinical dose of trastuzumab is 8 mg/kg administered three times a week (Q3W). The receptor occupancy (RO) at this dose is not available at this time. However, predicted data from a HER2 and HER3 dual antibody, MCLA-128, suggests that a flat dose of 480 mg, which is almost 8 mg/kg, would achieve 100% RO in clinical trials[143]. Assuming DAR (Drug-to-Antibody Ratio) is 2, even if only 1/10th of the antibody enters the tumor, the amount of payload that can be delivered into the tumor is significant.

The immune stimulatory components of ISAC target TLRs, which represent a promising approach in tumor immunotherapy. By activating TLRs, the immune system can be reactivated to combat cancer cells, making it a beneficial addition to cancer comprehensive therapy. However, systemic administration of TLR agonists has been associated with serious toxicity, which limits their clinical application to intra-tumoral administration only. The non-specific immune responses triggered by systemic administration can lead to cytokine storms, which can cause severe cases. Researchers are now focusing on developing small-molecule immunostimulants that are safer for systemic administration.

One such TLR8 agonist is motolimod (VTX-2337), which has an EC50 of about 100 nM for TLR8 activation and a similar EC50 for activation of PBMC secretory cytokines. It is also active against TLR7/8 and has a lower impact. In Phase I clinical trials, the Cmax of single-agent

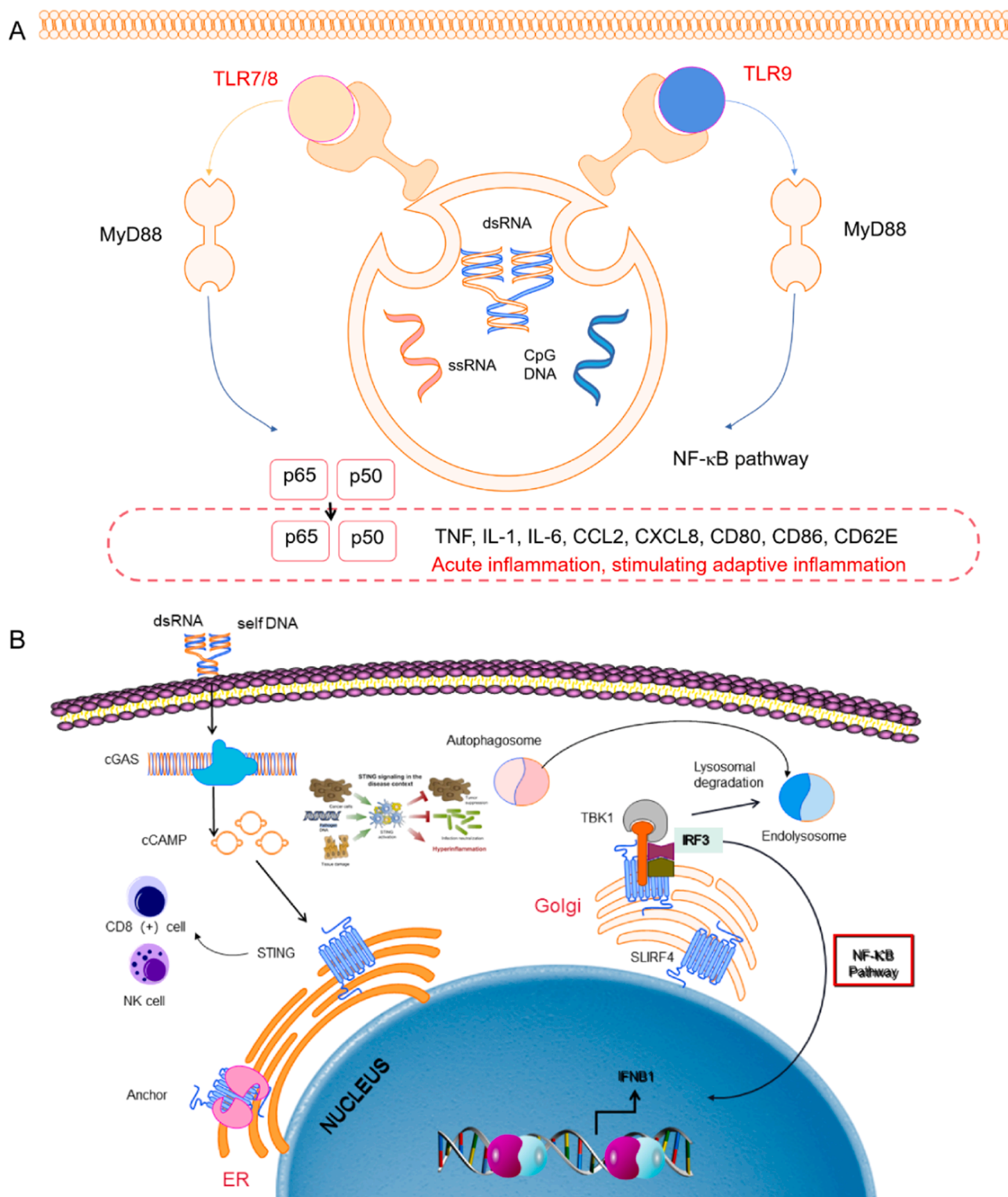


Fig. 2. TLR7/8, TLR9, and STING activate immune mechanisms. (A) TLR7/8, TLR9 activate immune mechanisms. (B) STING in the cell and activate immune mechanisms.

motolimod was 3.9 mg/m², resulting in a C_{max} of about 23 ng/ml, with a molar concentration of about 50 nM, which is lower than the EC₅₀ observed in vitro [144].

Another TLR7/8 agonist, Telratolimod (MEDI9197), is more active and triggers macrophage activation more effectively than LPS in vitro. It activates DCs to secrete INF and plateaus at 100 nM. The clinical safety window is narrow, with intra-tumoral administration once every four

weeks and an MTD of only 0.037 mg, which is similar to IL2 in terms of tolerability [145]. The main dose-limiting toxicity (DLT) is cytokine release syndrome. There is currently no information available about local exposure or PK. Assuming that all of these drugs are absorbed and target the tumor, the accumulation dosage can reach tens of nanomoles, which should be higher than that of VTX-2337. However, TLR-mediated immune activation in vivo is a positive feedback system, making it

Table 2
STING agonists for cancer patients in clinical trials.

Target (s)	Agonist	Indication(s)	Phase (s)	Status	Route	Notes	Clinical Number	Ref
STING	ADU-S100	Metastatic Head and Neck Cancer	II	Terminated	i.v.	Combined with Pembrolizumab	NCT03937141	[92]
		Advanced/Metastatic Solid Tumors or Lymphomas	I	Terminated	i.t.	Combined with Ipilimumab	NCT02675439	[93]
	MK-1454	Solid Tumors and Lymphomas	I	Terminated	i.v.	Combined with PDR001	NCT03172936	[94]
		Head and Neck Squamous Cell Carcinoma (HNSCC)	II	Completed	i.v.	Combined with Pembrolizumab	NCT04220866	[95]
		Solid Tumors Lymphoma	I	Completed	i.v.	As a single agent	NCT03010176	[96]
	BMS-986301	Advanced Solid Cancers	I	Recruiting	i.m.	As a single agent	NCT03956680	[97]
	TAK-676	Carcinoma, Non-Small-Cell Lung	I	Recruiting	i.v.	Combined with Pembrolizumab	NCT04879849	[98]
		Triple Negative Breast Neoplasms						
		Squamous Cell Carcinoma of Head and Neck						
		Solid Tumor	I	Recruiting	Intratumoral microdose injection	Combined with Carboplatin , 5-FU , Paclitaxel	NCT04541108	[99]
	E7766	Lymphoma	I	Recruiting	intratumorally	As a single agent	NCT04144140	[100]
		Advanced Solid Tumors						
	GSK3745417	Urinary Bladder Neoplasms	I	Withdrawn	i.v.	As a single agent	NCT04109092	[101]
		Leukemia, Myeloid, Acute	I	Recruiting	i.v.	As a single agent	NCT05424380	[102]
		Neoplasms	I	Recruiting	i.v.	Combined with Dostarlimab	NCT03843359	[103]
	SB11285	Melanoma	I	Recruiting	i.v.	Combination With Atezolizumab	NCT04096638	[104]
		Head and Neck Squamous Cell Carcinoma						
		Solid Tumor						
	IMSA101 (GB492)	Solid Tumor	I/II	Recruiting	i.v.	As a single agent	NCT04020185	[105]
	HG381	Advanced Solid Tumor	I	Recruiting	i.v.	As a single agent	NCT04998422	[106]

challenging to use preclinical data as a reliable indicator of effective dosing.

Based on these mechanisms, key elements of ISAC design include target selection as the starting point for the design of ISACs, determining which tumor markers the ISAC will target, and potentially influencing the choice of drug to be conjugated. The choice of antibody type is more likely to be based on IgG1. Antibody design requires a combination of factors such as affinity, internalization efficiency, and other indicators with no obvious shortcomings. Only a few immunostimulants are currently used in the design of ISAC, and they must possess critical properties beyond toxicity and immunostimulatory response, such as a balance of affinity and good stability. The ISAC linker must ensure that the drug remains intact while circulating in the blood vessels and that it is released once inside the tumor cells, to reduce side effects while ensuring efficacy.

4. ISAC existing problems and possible solutions

4.1. The existing question of ISAC

Early on in the BDC-1001 preclinical study, researchers tested two agonists, CL264 and T785, and found that agonist potency affected the efficacy of ISAC. According to the results, CL264-ISAC was more potent than T785-ISAC in inducing myeloid APC activation and antitumor activity in vivo, but the former resulted in systemic cytokine secretion and transient weight loss, suggesting possible associated toxic side effects. This may be why BDC-1001 chose the less active agonist in the clinical trial. Reviewing the clinical failures of both companies it is also easy to see that the problem lies in the choice of agonist. Agonists who were too active could cause unbearable side effects, while agonists who were too inactive could reduce side effects but have little efficacy. The ideal ISAC requires a balance of toxicity and efficacy. Choosing a suitable agonist that maintains efficacy while controlling toxicity is a major research and development challenge for ISACs. At present, ISACs seem to have difficulty in crossing the dose creep hurdle. For agonists, the increase in

efficacy will be accompanied by an increase in toxicity, and it is not easy to increase efficacy while maintaining safety.

SBT6050, the HER2 monoclonal antibody is linked to the TLR8 agonist, using a cleavable linker. The EC₅₀ of the payload is unknown, the data that we can see is that the EC₅₀ of the activated myeloid cells after good coupling is between 0.1 and 1 nM, showing a great effect [146]. The antibody binds HER2 on the tumor cells and then the myeloid cells snatch the drug up and swallow it by the FcR, but how does it get snatched up by the myeloid cells? If this mechanism is correct, the ISAC will kill all the cells that have FcR? So, we also have to consider the endocytosis of HER2. HER2 is a poor target for endocytosis, someone previously tested the endocytosis rate of Herceptin, about 48 hours, after the relentless efforts of tumor cells to endocytose less than 20% [132,133]. This shows that the efficacy of the drug is not directly related to the speed of endocytosis. It depends on how many of these cells endocytose, HER2-overexpressing cancer cells have millions of copies number on the surface. In short, SBT6050 will be swallowed by both myeloid cells and tumor cells and because of the HER2 monoclonal antibody on the surface of tumor cells, it will also be swallowed by macrophages through ADCP (antibody-dependent cellular phagocytosis) effect, which will release payload after swallowing [147]. The maximum tolerated dose and RP2D is 0.6 mg/kg, and Silverback believes that the target saturation is reached at 0.6 mg/kg. However, HER2 receptor occupancy (RO) trastuzumab 6–8mpk can reach 100% and 0.6mpk is far from the targeting effect [148,149]. This data means that the dose of TLR8 agonist is enough to exert the function whereas the dose of HER2 monoclonal antibody is not enough to target. In fact, in the preclinical data for SBT6050, when the dose was 10 mg/kg, the efficacy was good but the side effects were also greater, and allergic reactions occurred. SBT6050 had to be reduced and the dosing method changed to subcutaneous injection. The side effects persisted after the dose was reduced. In clinical trials, SBT6050 doses up to 1.2 mg/kg Q2W already showed grade 3 DLT (dose limiting toxicity), making it difficult to continue increasing its dose, further hard to improve efficacy. The difference is due to the difference in agonist activity [119,147]. The agonist

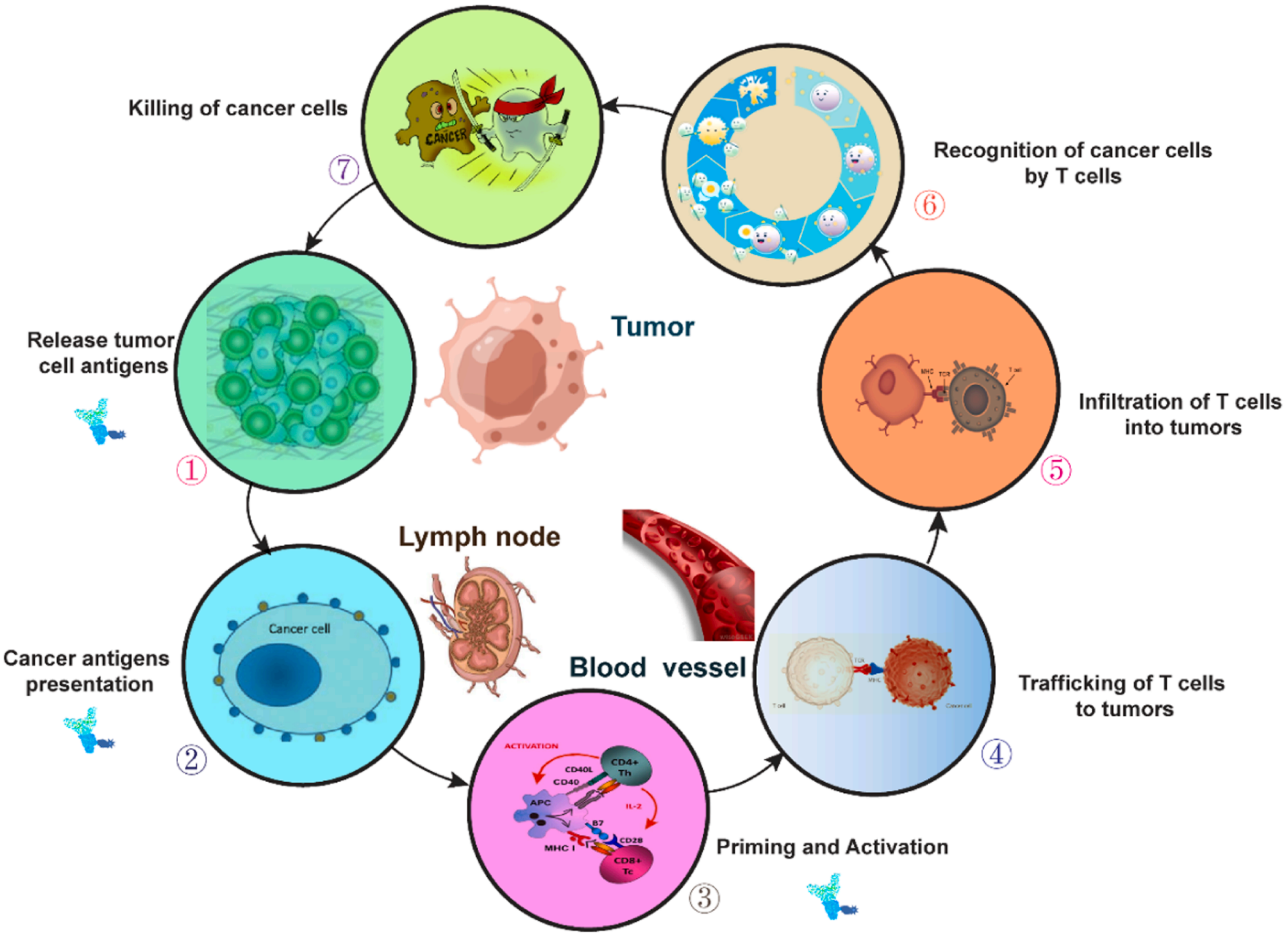


Fig. 3. ISACs initiate an entirely new immune response. (1). Tumor antigen release; (2). Tumor antigen presentation; (3). Priming and activation of effector T-cells; (4). T-cell migration to tumor tissue; (5). T-cell infiltration in tumor tissue; (6). T-cells recognize tumor cells; (7). Remove tumor cells.

Table 3
ISAC in clinical trials.

Company	Drug	Payload	Target	Development phase	Clinical Number	Ref
Bolt	BDC-1001	TLR7/8(T785)	HER2	Phase I/II	NCT04278144	[111,112]
	BDC-2034	TLR7/8(T785)	CEA	IND		[113]
	4521/11	TLR7/8	PD-L1	Pre-Clinical		[114]
	Myeloid Modular(BDC-3042)	None	Dectin-2	Pre-Clinical		[115]
Immunosensor	IMSA101(GB492)	STING	PD-L1	Phase I/II	NCT04020185	[116]
	Silverback	TLR8	HER2	Terminated/ Active, not recruiting	NCT05091528	[117–119]
					NCT04460456	
				Terminated	NCT05234606	
Tallac	SBT6290	TLR8	Nectin4	Phase I/II	NCT05399654	[120]
Takeda	TAC-001	TLR9	CD22	Phase I/II	NCT05070247	[121]
Mersana	XMT-2056	STING	PD-1	Phase I	NCT05070247	[122]
Sutro Biopharma	STRO-001	STING	HER2	Phase 1	NCT05514717	[123]
	STRO-002	TLR7/8	CD74	Phase I	NCT03424603	[124,125]
		TLR7/8	FolRα	Phase I	NCT05200364	
					NCT03748186	
Ambrx Biopharma	CC-99712	TLR7/8	BCMA	Phase I	NCT04036461	[128]
	ARX-788	TLR7	HER2	Phase I/II	NCT04829604	[129,130]
					NCT03255070	
					NCT05018676	

selected for BDC-1001 was much less active, so the dose creeps to 12 mg/kg Q2W did not result in a DLT, but also did not result in a good clinical response.

As pioneers in the field of ISAC, the failure of these two companies has left the future of ISAC in limbo. ISAC can simultaneously activate both innate and adaptive immunity. While most immunotherapies currently focus on adaptive immunity, ISAC has the unique capability to activate both types of immunity, resulting in a more potent immune

response. However, the simultaneous activation of both immune responses carries an increased risk. This systemic immune activation can lead to non-specific attacks by the immune system, potentially causing harm to the body. In terms of overall development progress, ISAC is in a very early stage. Although ISAC has a clear concept, it faces difficulties in overcoming the dose escalation hurdle in clinical trials. The development of TLRs, STING, and other agonists is often accompanied by the challenges of narrow therapeutic windows and systemic toxicity. While

efficacy may increase, toxicity also tends to rise. Compared to ADC, ISAC has a more challenging therapeutic window to control.

Another limitation is the production of anti-drug antibodies, which ISAC may promote the production of anti-drug antibodies and affect the efficacy of drugs. ISAC achieves tumor suppression through tumor antigen recognition, Fc-mediated endocytosis of APC, and TLR receptor activation in APCs. Ideal ISAC drives innate and specific immunity and generates immune memory, activates dendritic cells and other APC, activates CD4/CD8 T cells by TLR, where cold tumors become hot tumors, and can be used in combination with other immune checkpoint drugs. However, more human clinical validation is needed to determine whether the human autoimmune system alone is sufficient to complete human tumor clearance.

4.2. Solutions to ISAC development constraints

Dual-target, dual-mechanism drugs need a balance between efficacy and safety. The ideal situation is to find a dose that maximizes the efficacy of both mechanisms and is still in the safety window, the so-called 'sweet spot'.

Because the TLR7/8 agonist is too toxic, we can reduce the activity of the payload. The ideal situation would be to administer a dose of 6–8 mpk, where the HER2 monoclonal antibody is fully functional, and the TLR agonist end of the dose reaches the target saturation, while the patient develops a tolerable TLR agonist typical AE (adverse event), such as fever, headache, chills, however it with a therapeutic effect. The strategy utilized by BOLT Therapeutics involves activating myeloid cells to enhance adaptive immune responses for killing tumors. This approach aims to transform "cold" tumors into immunologically "hot" tumors, thereby exposing them to the immune system for recognition and elimination by tumor-killing cells. By tackling the challenge of low response rates to single immune checkpoint inhibitors, this strategy aims to enhance the effectiveness of immunotherapy. Its BDC-1001 uses a low-activity TLR7/8 agonist and a non-cleavable linker so that the overall activity of the TLR side is reduced considerably. The dose of trastuzumab is 6–8 mpk Q3W to HER2 100% RO, then BDC-1001 climbs to 20 mpk QW, usually, ISAC relies on HER2 targeting to enrich the payload, 8 mpk trastuzumab HER2 can reach 100% RO, the rising dose has not reached the expected effect, so reduce the more toxic. The activity of this arm cannot be reduced too much either. At this time, DAR can be adjusted to achieve the goal. DAR has an impact on drug pharmacology and activity and is essential for the later stages of ISAC development. ISAC is ingested by tumor cells in limited amounts during circulation in vivo, so generally higher DAR is beneficial for increased potency [150]. However, the immunostimulant used in ISAC has strong hydrophobicity, and a high DAR value will cause immunostimulant aggregation, leading to a decrease in the half-life of circulation in vivo, systemic immune response and side effects, which leads to high DAR are not desirable. The DAR value of preclinical and clinical ADCs is generally in the range of 2–8. If ISAC replaces the linker with a cleavable one, could this end of the TLR be made more potent? It would be better to get the optimal dose to 6–8 mpk. ADC like the new generation wonder drug DS8201 [151], which can be considered a dual mechanism drug, has a clinical dose of 5.4–6.4 mpk Q3W, which is simply a sweet spot paradigm. As mentioned above, it can be considered a viable strategy to achieve a balance between the two ends of ISAC by adjusting DAR.

The dual mechanism drug and the sweet spot theory are described. Another example is the PD-L1×4-1BB dual antibody. It binds PD-L1 positive tumor cells in one arm and 4-1BB positive T cells in the other arm, forming a tumor cell-antibody-T cell trimer, the formation of which is necessary for 4-1BB activation and is one of the sources of the dual antibody's potency. The second source of potency is the blockade of PD-L1 and PD1 pooling, which is visualized by the receptor occupancy of PD-L1. For PD-L1×4-1BB dual antibodies, the output of maximal anti-tumor activity is that the maximum rate of trimer formation is achieved while the receptor occupancy of PD-L1 is also 100%. The sweet spot dose

is the dose that achieves both maximum trimer formation and maximum PD-L1 receptor occupancy without causing hepatotoxicity of 4-1BB. Primary resistance to other PDL1-blocking antibodies is often caused by cold tumors and acquired resistance is often caused by T-cell depletion. 4-1BB (CD137) is an immune co-stimulatory target that is expressed on activated T-cells after T-cell activation [152]. It is also expressed on depleted T cells, and activation of 4-1BB can reverse T cell depletion. The first generation of the 4-1BB monoclonal antibody can activate 4-1BB directly without cross-linking, but the efficacy of the drug is seen, there are 4-1BB positive cells in the liver, resulting in hepatotoxicity when the drug climbs the slope. The second phase dose was lowered and the drug became ineffective again (4-1BB alone cannot be activated and requires the Fc-terminus to be captured by something else, such as FcR-positive cells, to be activated) However, the clinical efficacy of cross-linking-dependent antibodies is not much better, and we speculate the cross-linking ability of FcR alone is not quite sufficient [153].

Two of the more clinically advanced dual antibodies are Genmab's GEN1046, a 1+1 bivalent antibody [154], and Inbrx's INBRX105, a 2+2 quadrivalent antibody [155]. The QSP (Quantitative Systems Pharmacology) semi-mechanical simulation shows that GEN1046 can achieve maximum trimer formation and >90% PD-L1 receptor occupancy at a dose of 1–2 mg/kg and that this dose does not cause liver toxicity, making it possible to achieve the sweet spot. GEN1046 of Genmab. Because it's bivalent, easily competing with itself for the hook effect, both Deqi and Genmab ran QSP simulations on GEN1046 and got similar results. When Genmab reached the highest dose of the trimer, the occupancy of the PD-L1 receptor was far from full, but the dose continued to rise, and the occupancy of the PD-L1 receptor was increased. However, because of the monovalent binding to both targets, the number of trimers formed decreased with the increase in dose, showing a hook effect. Genmab's PD-L1 receptor occupancy was far from full at the highest dose of trimerization, however as the dose continued to rise, PD-L1 receptor occupancy went up because of the monovalent binding to both targets, the dose rose, GEN1046 could not find a dose where trimmer was high and PD-L1 RO was high. This simulated result is also consistent with Genmab's clinical data, where GEN1046 Phase I climbed to 1200 mg flat dose without reaching MTD, but they chose a 100 mg dose in Phase II, also because they saw a hook effect [156]. A very reasonable design uses a PD-L1×4-1BB double antibody, using PD-L1 positive tumor cells to enrich the antibody and cross-link it to activate 4-1BB, also a double mechanism: one foot on the oil, one foot off the brake. This can also be applied to ISAC, to balance the relationship between the effects of antibodies and immune stimulants.

The TCR mentioned above can activate immunity, in actuality, the B-cell receptor (BCR) can also activate intrinsic immunity. BTK is a non-receptor tyrosine kinase that links BCR signaling, TLR signaling, and chemokine receptor signaling [157]. While BCR signaling drives the proliferation of malignant B cells, such as chronic lymphocytic leukemia (CLL) cells, TLR, and chemokine receptor signaling are highly associated with inflammation, making BTK inhibitors a potential therapeutic agent for COVID-19. Over the past decade, BTK inhibitors have been increasingly used as an alternative to conventional chemotherapy approaches, particularly in patients with chronic lymphocytic leukemia (CLL) and mantle cell lymphoma (MCL). The over-reactive immune response to CRS caused by SARS-CoV-2 in patients with neocoronary pneumonia has facilitated clinical studies using BTK inhibitors in the treatment of patients with moderate to severe neocoronary pneumonia [158]. These studies have shown promising results for BTK inhibitors in reducing the severity and mortality of the immune response due to infection. The immune activator component of ISAC could be replaced with a BTK inhibitor regulating immune activation activity in future ISAC development.

During the stage of new drug development, the targeted ability based on the antibody portion is the most prominent feature and advantage of ISAC compared to traditional chemical drugs. Generally, the higher the affinity, the more precise the antibody recognition. However, Some

researches have also indicated that antibodies with stronger affinity as immune stimulants may have limited penetration effects on solid tumors [159–162]. Therefore, in the selection of antibodies, higher affinity does not necessarily equate to better outcomes; similarly, when antibody affinity is too low, drug effectiveness may not be optimal. Hence, in the selection or construction of antibodies, attention should be paid to both the penetration ability and affinity of the antibodies, while balancing their relationship and DAR. During the clinical trial phase, it is crucial to consider the immune-related adverse event (irAE), the specificity caused by different categories of immune stimulants, the individual differences in the tumor immune microenvironment exacerbating irAEs, as well as factors such as prior immunotherapy use, involvement of unknown genetic backgrounds, and the potential for specific irAEs induced by different organ-specific immune microenvironments.

5. Conclusions

With the recent setbacks in the development of ISACs, it is too early to dismiss them as a failed concept. The process of turning scientific ideas into reality is complex and often involves many obstacles and setbacks. Although the antibody and linker technologies used in ISACs are relatively mature, the replacement of the load with immune agonists or modulators is a significant challenge. The efficacy of ISACs can be affected by many factors, such as the selection of agonists, binding epitopes, valency, and antibody affinity. The development of ISACs requires a careful balance of both efficacy and safety, as the drugs have dual mechanisms of action. The design principle is: “Heaven takes away from those who have too much, and gives to those that have not enough”. Overall, ISACs have great potential in targeted therapeutics, but further research and development are required to optimize their effectiveness and overcome their limitations. More consideration of inter-individual variability of the immune microenvironment in clinical applications. This review offers a comprehensive overview of ISACs, emphasizing their potential in treating cancer and autoimmune diseases, and discussing strategies to enhance their design and bioavailability. The development of ISAC requires more conceptual and clinical validation, making the exploration path arduous and long. As special ADCs, ISACs are expected to lead the new generation of ADCs development. “Balance” can make ISAC go further and further.

Ethical Approval

Not applicable

Authors' contributions

All authors have read and approved the article. Concept and design: C.F., W.W.T. Drafting of the manuscript: C.F., W.W.T., L.F.Y., Q.W., Y.X.M., and Z.J.Y. Revision of the manuscript for important intellectual content: Z.J.Y., B.C., and M.J.W. Obtained funding: M.J.W., L.F.Y., Z.J.Y., and C.F. Supervision and Corresponding authors: Z.J.Y., B.C., and M.J.W.

Consent for publication

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CRedit authorship contribution statement

Yuxi Miao: Writing – review & editing, Visualization, Project administration, Investigation, Data curation. **Qian Wei:** Validation, Supervision, Resources, Project administration, Investigation, Formal analysis. **Lifeng Yu:** Visualization, Project administration, Methodology, Funding acquisition, Data curation. **Chen Fu:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Weiwei Tong:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Minjie Wei:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation. **Zhaojin Yu:** Writing – original draft, Visualization, Resources, Funding acquisition, Formal analysis, Conceptualization. **Bo Chen:** Writing – review & editing, Supervision, Project administration, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service, and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled “When will the immune-stimulating antibody conjugates (ISACs) from bench to bedside?”.

Availability of data and materials

Not applicable

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